

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: import warnings
warnings.filterwarnings("ignore")
```

```
In [3]: df = pd.read_csv("walmart_data.csv")
```

```
In [4]: df.head()
```

Out[4]:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	M
0	1000001	P00069042	F	0-17	10	A	2	
1	1000001	P00248942	F	0-17	10	A	2	
2	1000001	P00087842	F	0-17	10	A	2	
3	1000001	P00085442	F	0-17	10	A	2	
4	1000002	P00285442	M	55+	16	C	4+	

1. Defining Problem Statement and Analyzing basic metrics .

1. Observations on shape of data, data types of all the attributes, conversion of categorical attributes to 'category' (If required), statistical summary

```
In [5]: df.shape

print(f"There are {df.shape[0]} rows and {df.shape[1]} columns are present in
```

There are 550068 rows and 10 columns are present in the dataset

```
In [6]: df.dtypes
```

```
Out[6]: User_ID                int64
Product_ID              object
Gender                  object
Age                    object
Occupation              int64
City_Category           object
Stay_In_Current_City_Years  object
Marital_Status          int64
Product_Category         int64
Purchase                int64
dtype: object
```

In [7]: `df.info(memory_usage='deep')`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   User_ID                             550068 non-null  int64
1   Product_ID                          550068 non-null  object
2   Gender                              550068 non-null  object
3   Age                                  550068 non-null  object
4   Occupation                          550068 non-null  int64
5   City_Category                       550068 non-null  object
6   Stay_In_Current_City_Years          550068 non-null  object
7   Marital_Status                      550068 non-null  int64
8   Product_Category                    550068 non-null  int64
9   Purchase                            550068 non-null  int64
dtypes: int64(5), object(5)
memory usage: 179.4 MB
```

In [8]: `df['Product_ID'] = df['Product_ID'].astype('category')`
`df["Gender"] = df["Gender"].astype("category")`
`df["Age"] = df["Age"].astype("category")`
`df["Occupation"] = df["Occupation"].astype("category")`
`df["Product_Category"] = df["Product_Category"].astype("category")`
`df["Marital_Status"] = df["Marital_Status"].astype("category")`
`df["City_Category"] = df["City_Category"].astype("category")`
`df["Stay_In_Current_City_Years"] = df["Stay_In_Current_City_Years"].astype("category")`
`df["User_ID"] = df["User_ID"].astype("category")`

In [9]: `df.describe()`

Out[9]:

	Purchase
count	550068.000000
mean	9263.968713
std	5023.065394
min	12.000000
25%	5823.000000
50%	8047.000000
75%	12054.000000
max	23961.000000

In [41]: `df.describe(include = "category")`

Out[41]:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_
count	550068	550068	550068	550068	550068	550068	550068
unique	5891	3631	2	7	21	3	3
top	1001680	P00265242	M	26-35	4	B	B
freq	1026	1880	414259	219587	72308	231173	231173

In [42]: `df.dtypes`

Out[42]:

User_ID	category
Product_ID	category
Gender	category
Age	category
Occupation	category
City_Category	category
Stay_In_Current_City_Years	category
Marital_Status	category
Product_Category	category
Purchase	int64
dtype:	object

In [43]: `df.info(memory_usage='deep')`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   User_ID                               550068 non-null  category
1   Product_ID                             550068 non-null  category
2   Gender                                 550068 non-null  category
3   Age                                    550068 non-null  category
4   Occupation                             550068 non-null  category
5   City_Category                           550068 non-null  category
6   Stay_In_Current_City_Years             550068 non-null  category
7   Marital_Status                         550068 non-null  category
8   Product_Category                       550068 non-null  category
9   Purchase                               550068 non-null  int64
dtypes: category(9), int64(1)
memory usage: 10.5 MB
```

2. Non-Graphical Analysis: Value counts and unique attributes

In [44]: `df["Gender"].value_counts()`

Out[44]:

M	414259
F	135809
Name: Gender, dtype: int64	

```
In [45]: df["Age"].value_counts()
```

```
Out[45]: 26-35    219587
          36-45    110013
          18-25     99660
          46-50     45701
          51-55     38501
          55+      21504
          0-17     15102
          Name: Age, dtype: int64
```

```
In [46]: df["City_Category"].value_counts()
```

```
Out[46]: B    231173
          C    171175
          A    147720
          Name: City_Category, dtype: int64
```

```
In [47]: df["Stay_In_Current_City_Years"].value_counts()
```

```
Out[47]: 1    193821
          2    101838
          3     95285
          4+    84726
          0     74398
          Name: Stay_In_Current_City_Years, dtype: int64
```

```
In [48]: df["Marital_Status"].value_counts()
```

```
Out[48]: 0    324731
          1    225337
          Name: Marital_Status, dtype: int64
```

```
In [49]: df["Occupation"].value_counts().sort_index()
```

```
Out[49]: 0    69638
          1    47426
          2    26588
          3    17650
          4    72308
          5    12177
          6    20355
          7    59133
          8     1546
          9     6291
          10   12930
          11   11586
          12   31179
          13    7728
          14   27309
          15   12165
          16   25371
          17   40043
          18    6622
          19    8461
          20   33562
          Name: Occupation, dtype: int64
```

```
In [50]: df["Product_Category"].value_counts().sort_index()
```

```
Out[50]: 1      140378
          2      23864
          3      20213
          4      11753
          5     150933
          6      20466
          7       3721
          8     113925
          9        410
         10       5125
         11     24287
         12       3947
         13      5549
         14      1523
         15      6290
         16      9828
         17       578
         18      3125
         19      1603
         20      2550
          Name: Product_Category, dtype: int64
```

```
In [51]: df["Product_ID"].value_counts().sort_values(ascending = False)
```

```
Out[51]: P00265242      1880
          P00025442      1615
          P00110742      1612
          P00112142      1562
          P00057642      1470
          ...
          P00065542         1
          P00107442         1
          P00135942         1
          P00065942         1
          P00231642         1
          Name: Product_ID, Length: 3631, dtype: int64
```

```
In [52]: df["Product_ID"].nunique()
```

```
Out[52]: 3631
```

3. Visual Analysis - Univariate & Bivariate

(a) Univariate Categorical Variables

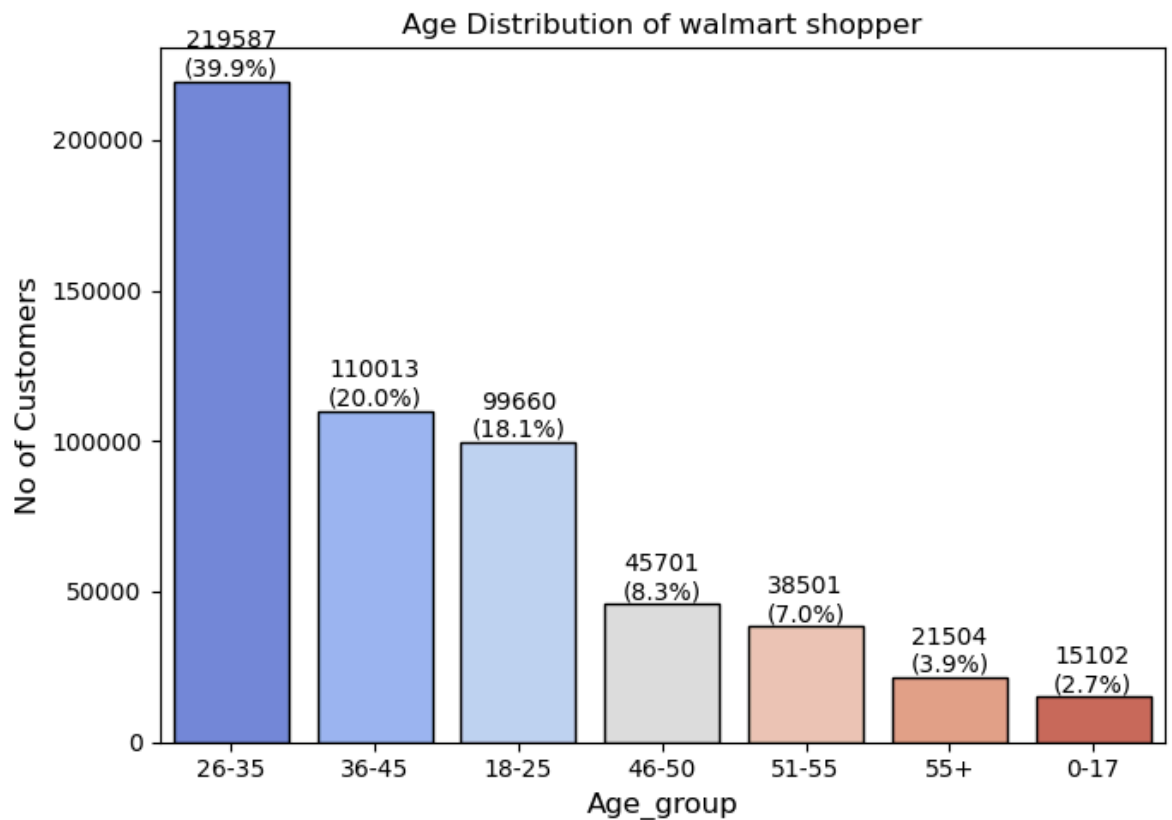
```
In [53]: plt.figure(figsize = (7, 5))  
sns.countplot(data = df, x= "Gender", palette="coolwarm",edgecolor="black")  
plt.xlabel("Gender", fontsize = 12)  
plt.ylabel("No of Customers", fontsize = 12)  
plt.title("Gender Distribution of walmart shopper")  
plt.show()
```



```
In [54]: plt.figure(figsize = (7, 5))
ax = sns.countplot(data = df, x= "Age",order=df['Age'].value_counts(ascending=
plt.xlabel("Age_group", fontsize = 12)
plt.ylabel("No of Customers", fontsize = 12)
plt.title("Age Distribution of walmart shopper")

# Add annotations
total = len(df)
for p in ax.patches:
    count = int(p.get_height())
    percentage = 100 * count / total
    ax.annotate(
        f'{count}\n({percentage:.1f}%)',          # text to display
        (p.get_x() + p.get_width() / 2., count),  # position of text
        ha='center', va='bottom',
        fontsize=10, color='black'
    )

# Improve layout
plt.tight_layout()
plt.show()
```



```
In [55]: plt.figure(figsize = (7, 5))
sns.countplot(data = df, x= "Occupation",order=df['Occupation'].value_counts(a
plt.xlabel("Occupation (Masked)", fontsize = 12)
plt.ylabel("No of Customers", fontsize = 12)
plt.title("Distribution of Walmart Shoppers by Occupation")
plt.xticks(rotation=45, fontsize=10)
plt.yticks(fontsize=10)
plt.show()
```




```
In [56]: sns.set(style="whitegrid", context="talk")
plt.figure(figsize = (7, 5))
ax = sns.countplot(data = df, x= "City_Category",order=df['City_Category'].value_counts().index)
plt.xlabel("City Category", fontsize = 12)
plt.ylabel("No of Customers", fontsize = 12)
plt.title("Distribution of Walmart Shoppers by City Category",fontsize=15, weight='bold')
plt.xticks(fontsize=11)
plt.yticks(fontsize=10)

# Add annotations
total = len(df)
for p in ax.patches:
    count = int(p.get_height())
    percentage = 100 * count / total
    ax.annotate(
        f'({percentage:.1f}%)',          # text to display
        (p.get_x() + p.get_width() / 2., count),  # position of text
        ha='center', va='bottom',
        fontsize=10, color='black'
    )
plt.show()
```



```
In [57]: df.columns
```

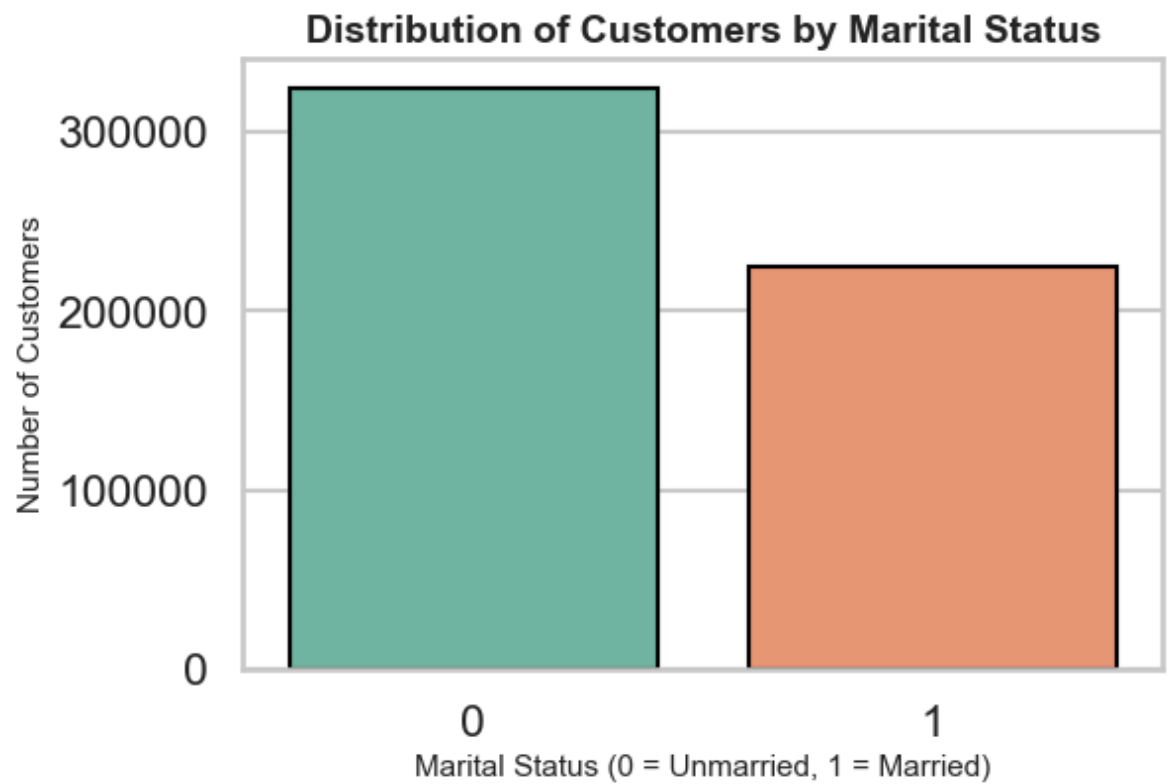
```
Out[57]: Index(['User_ID', 'Product_ID', 'Gender', 'Age', 'Occupation', 'City_Category',
                'Stay_In_Current_City_Years', 'Marital_Status', 'Product_Category',
                'Purchase'],
              dtype='object')
```

```
In [58]: sns.set(style="whitegrid", context="talk")
plt.figure(figsize = (7, 5))
ax = sns.countplot(data = df, x= "Stay_In_Current_City_Years",order=df['Stay_In_Current_City_Years'].unique(),
plt.xlabel("Years Spent in Current City", fontsize = 12)
plt.ylabel("No of Customers", fontsize = 12)
plt.title("Distribution of Walmart Shoppers by City Tenure",fontsize=15, weight='bold')
plt.xticks(fontsize=11)
plt.yticks(fontsize=10)

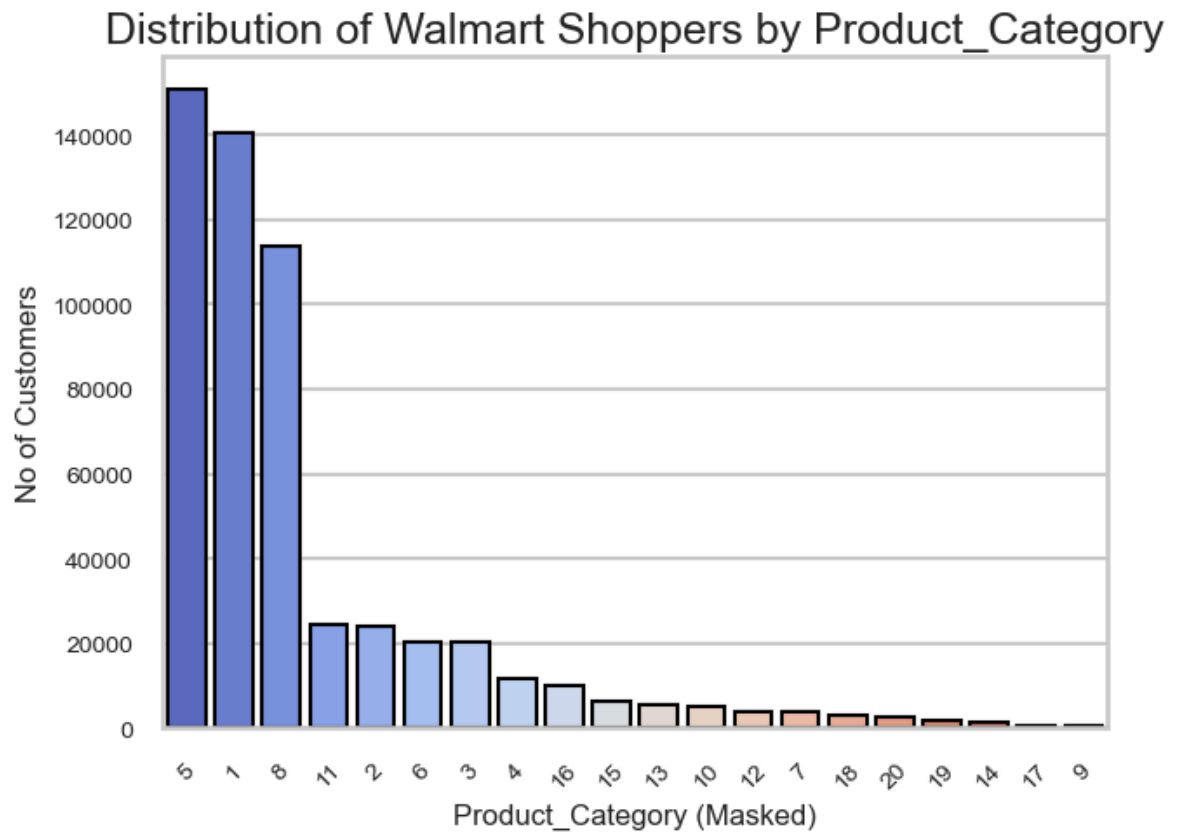
# Add annotations
total = len(df)
for p in ax.patches:
    count = int(p.get_height())
    percentage = 100 * count / total
    ax.annotate(
        f'({percentage:.1f}%)', # text to display
        (p.get_x() + p.get_width() / 2., count), # position of text
        ha='center', va='bottom',
        fontsize=10, color='black'
    )
plt.show()
```



```
In [59]: plt.figure(figsize=(6,4))
sns.countplot(data=df, x='Marital_Status', palette='Set2', edgecolor='black')
plt.title('Distribution of Customers by Marital Status', fontsize=14, weight='bold')
plt.xlabel('Marital Status (0 = Unmarried, 1 = Married)', fontsize=11)
plt.ylabel('Number of Customers', fontsize=11)
plt.show()
```

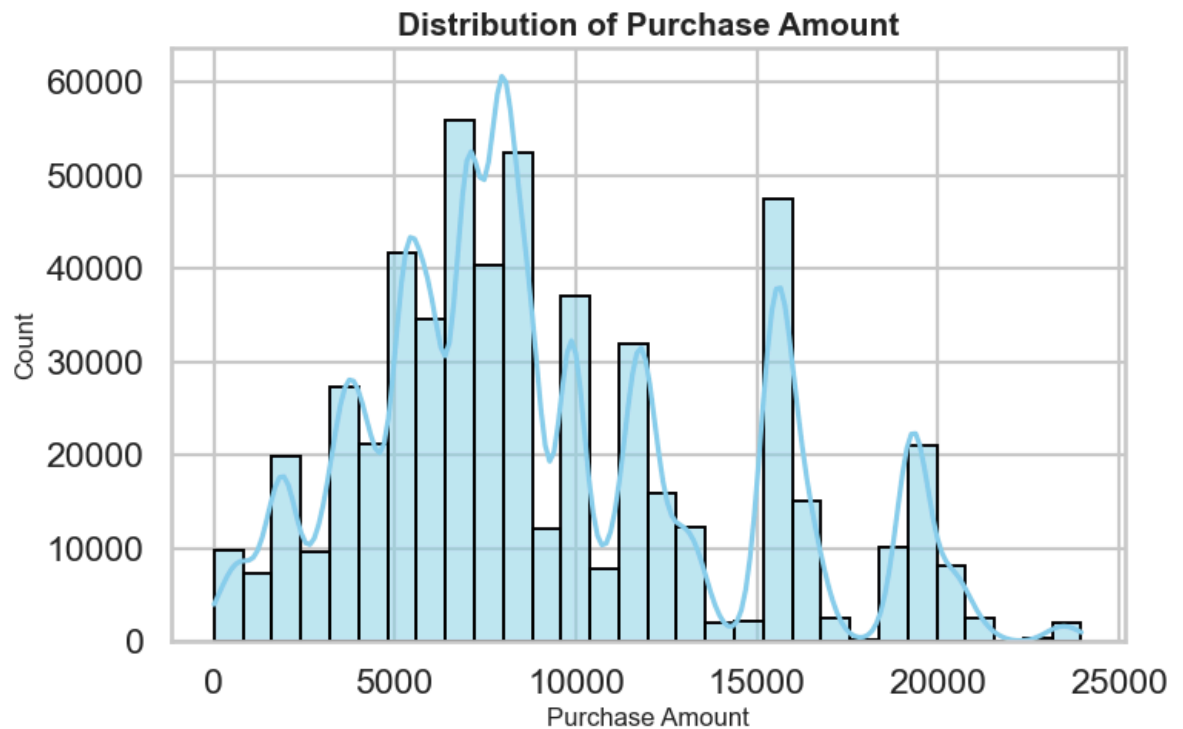


```
In [60]: plt.figure(figsize = (7, 5))
sns.countplot(data = df, x= "Product_Category",order=df['Product_Category'].va
plt.xlabel("Product_Category (Masked)", fontsize = 12)
plt.ylabel("No of Customers", fontsize = 12)
plt.title("Distribution of Walmart Shoppers by Product_Category")
plt.xticks(rotation=45, fontsize=10)
plt.yticks(fontsize=10)
plt.show()
```



(b) Continuous Variables — Purchase

```
In [61]: plt.figure(figsize=(8,5))
sns.histplot(df['Purchase'], kde=True, bins=30, color='skyblue', edgecolor='b')
plt.title('Distribution of Purchase Amount', fontsize=15, weight='bold')
plt.xlabel('Purchase Amount', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.show()
```



```
In [62]: df["Purchase"].skew()
```

```
Out[62]: 0.6001400037087128
```

- The distribution is slightly positively skewed, meaning the tail is longer on the right side. This indicates that most purchases are below the average, but a smaller number of high-value purchases pull the mean up.

```
In [63]: df["Purchase"].kurt()
```

```
Out[63]: -0.3383775655851702
```

- The distribution is slightly platykurtic, meaning it has lighter tails than a normal distribution.

Sampling Distribution of Mean Purchase

```

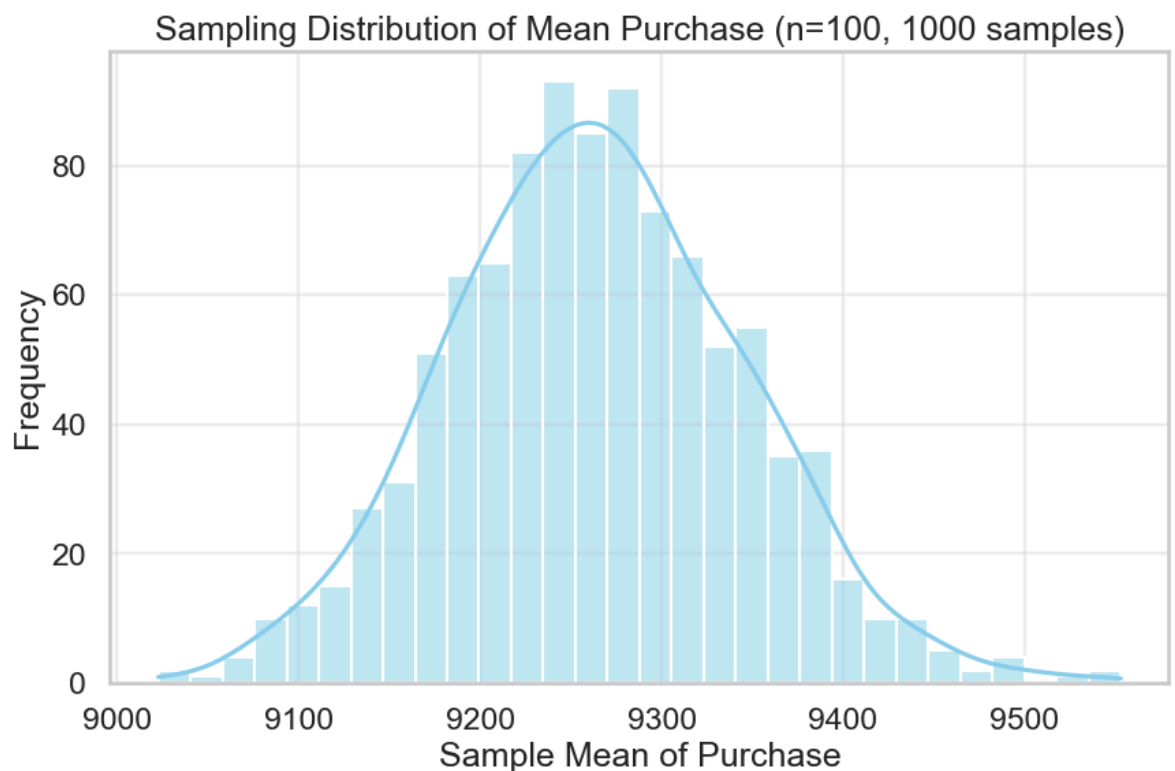
In [135]: sample_size = 4000      # n = 4000 per sample
          num_samples = 1000     # number of samples to draw

          sample_means = []

          for i in range(num_samples):
              sample = np.random.choice(df['Purchase'], size=sample_size, replace=True)
              sample_means.append(np.mean(sample))

          plt.figure(figsize=(10,6))
          sns.histplot(sample_means, kde=True, color='skyblue', bins=30)
          plt.title("Sampling Distribution of Mean Purchase (n=100, 1000 samples)")
          plt.xlabel("Sample Mean of Purchase")
          plt.ylabel("Frequency")
          plt.grid(alpha=0.3)
          plt.show()

```



```

In [136]: sampling_distribution_mean = np.mean(sample_means)
          sampling_distribution_mean

```

Out[136]: 9263.570981750001

```

In [137]: np.mean(df["Purchase"])

```

Out[137]: 9263.968712959126

- The shape of the sampling distribution is approximately normal — even though original Purchase data was right-skewed.
- The mean of this sampling distribution $\approx$ the population mean of Purchase.
- The spread (Standard Error) decreases as the sample size (n) increases.

Bivariate Analysis

In [64]: `df.columns`

Out[64]: Index(['User_ID', 'Product_ID', 'Gender', 'Age', 'Occupation', 'City_Category',
 'Stay_In_Current_City_Years', 'Marital_Status', 'Product_Category',
 'Purchase'],
 dtype='object')

Purchase Amount by Gender

- Null Hypothesis (H_0): there is no significant difference between male and female spending
- Alternative Hypothesis (H_1): there is a significant difference between male and female spending

In [143]: `df["Gender"].value_counts()`

Out[143]: M 414259
 F 135809
 Name: Gender, dtype: int64

In [65]: `df.groupby("Gender")["Purchase"].mean()`

Out[65]: Gender
 F 8734.565765
 M 9437.526040
 Name: Purchase, dtype: float64

In [66]: `male = df[df["Gender"] == "M"]["Purchase"]
 female = df[df["Gender"] == "F"]["Purchase"]`

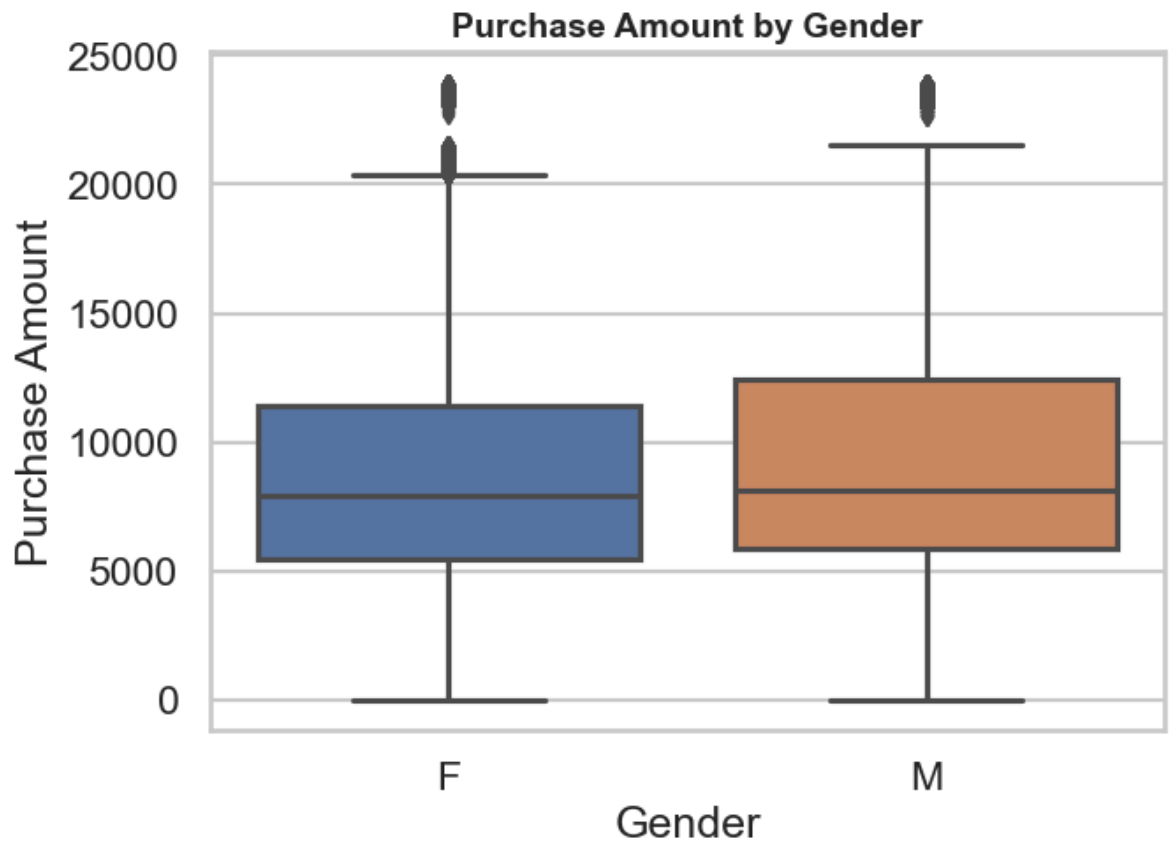
In [67]: `from scipy.stats import ttest_ind`

In [68]: `t, p = ttest_ind(male, female, alternative = "two-sided")
 t, p`

Out[68]: (44.837957934353966, 0.0)

In [69]: `if p < 0.05:
 print("there is a significant difference between male and female spending")
 else:
 print("there is no significant difference between male and female spending")`
 there is a significant difference between male and female spending

```
In [70]: plt.figure(figsize=(7,5))
sns.boxplot(data=df, x='Gender', y='Purchase')
plt.title('Purchase Amount by Gender', fontsize=14, weight='bold')
plt.xlabel('Gender')
plt.ylabel('Purchase Amount')
plt.show()
```



Purchase Amount by Age Group

```
In [139]: df["Age"].value_counts().sort_index(ascending = True)
```

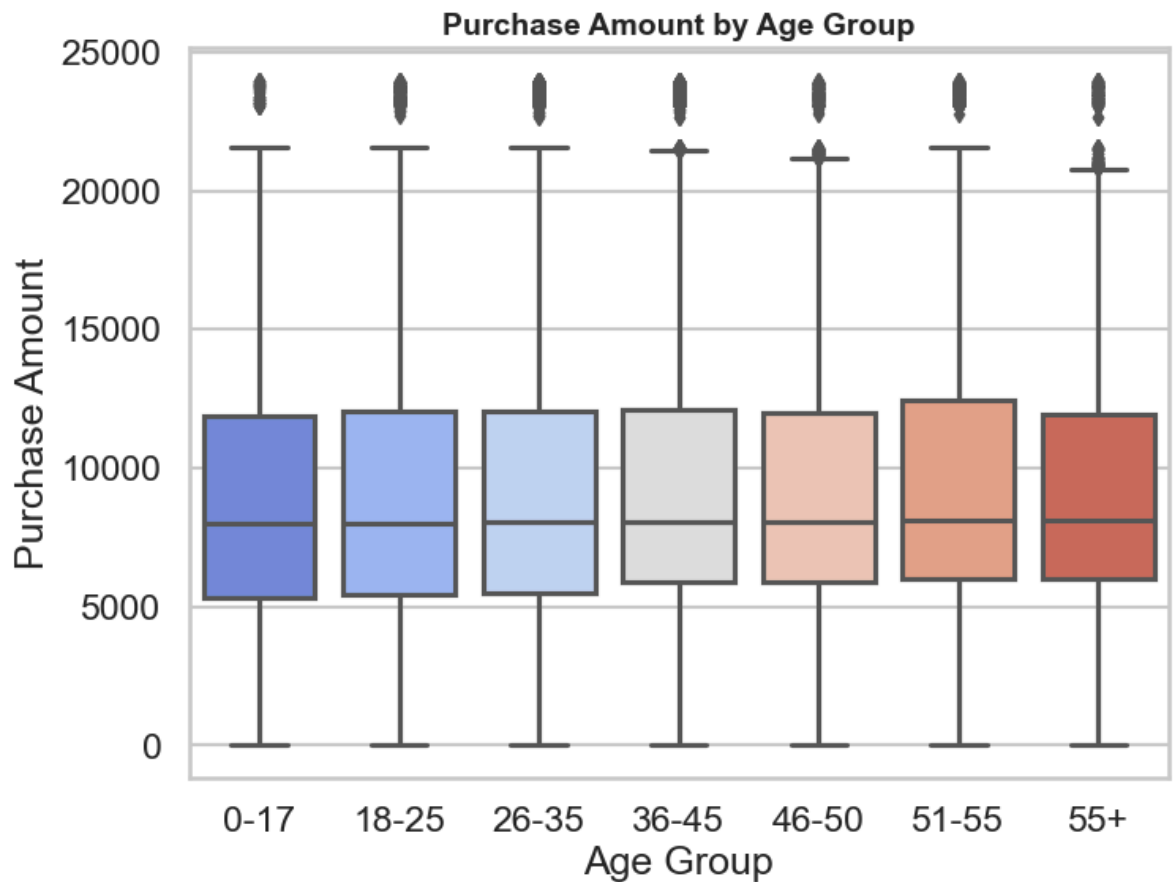
```
Out[139]: 0-17      15102
18-25      99660
26-35     219587
36-45     110013
46-50      45701
51-55      38501
55+       21504
Name: Age, dtype: int64
```

```
In [71]: df.groupby("Age")["Purchase"].mean()
```

```
Out[71]: Age
0-17      8933.464640
18-25     9169.663606
26-35     9252.690633
36-45     9331.350695
46-50     9208.625697
51-55     9534.808031
55+       9336.280459
Name: Purchase, dtype: float64
```



```
In [72]: plt.figure(figsize=(8,6))
sns.boxplot(data=df, x='Age', y='Purchase', palette='coolwarm')
plt.title('Purchase Amount by Age Group', fontsize=14, weight='bold')
plt.xlabel('Age Group')
plt.ylabel('Purchase Amount')
plt.show()
```



Purchase Amount by City Category

```
In [73]: df.groupby("City_Category")["Purchase"].mean()
```

```
Out[73]: City_Category
A      8911.939216
B      9151.300563
C      9719.920993
Name: Purchase, dtype: float64
```

- Null Hypothesis (H_0): There is no significant relation between city category and purchase
- Alternative Hypothesis (H_1): There is a significant relation between city category and purchase

```
In [74]: from scipy.stats import f_oneway
```

```
In [75]: A = df[df["City_Category"] == "A"]["Purchase"]
        B = df[df["City_Category"] == "B"]["Purchase"]
        C = df[df["City_Category"] == "C"]["Purchase"]
```

```
In [76]: f, p = f_oneway(A, B, C)
        f, p
```

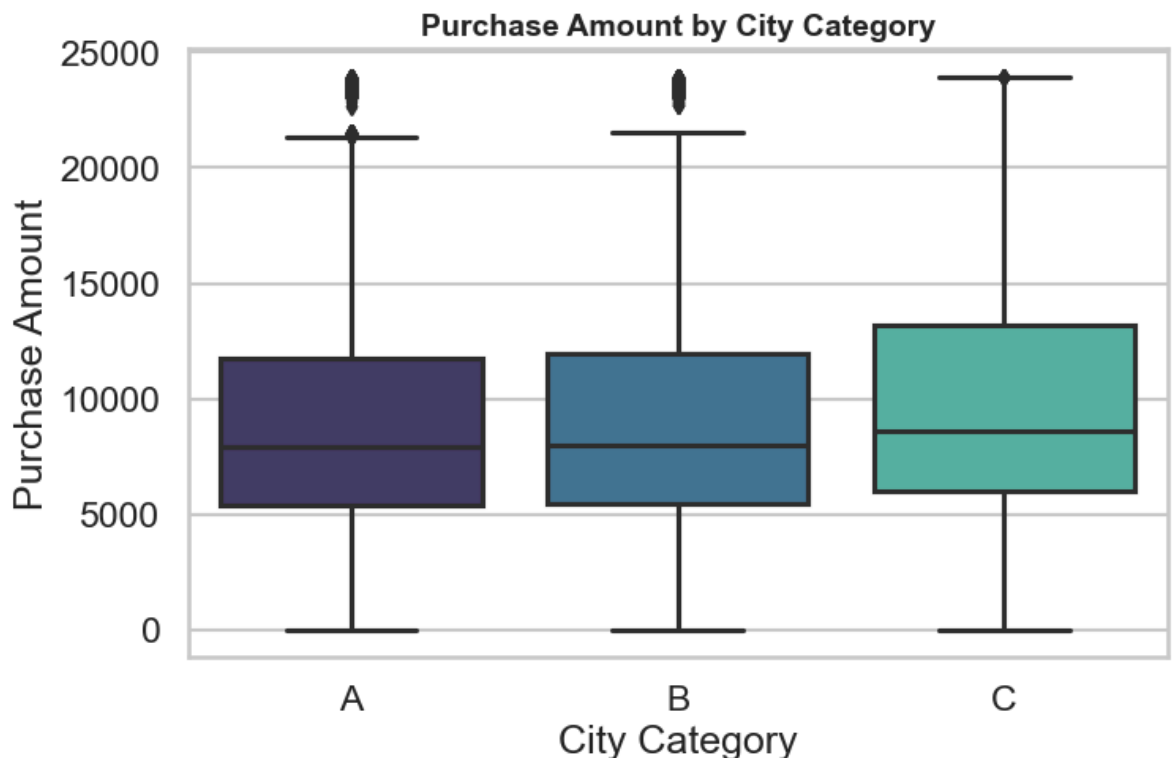
```
Out[76]: (1130.7459610102974, 0.0)
```

```
In [77]: if p < 0.05:
        print("There is a significant relation between city category and purchase")
    else:
        print("there is no significant relation between city category and purchase")
```

There is a significant relation between city category and purchase

city category significantly influences purchase behavior.

```
In [78]: plt.figure(figsize=(8,5))
        sns.boxplot(data=df, x='City_Category', y='Purchase', palette='mako')
        plt.title('Purchase Amount by City Category', fontsize=14, weight='bold')
        plt.xlabel('City Category')
        plt.ylabel('Purchase Amount')
        plt.show()
```



Purchase Amount by Marital Status

- Null Hypothesis (H_0): marital status has no significant impact on the average transaction value.
- Alternative Hypothesis (H_1): marital status has no significant impact on the average transaction value.

```
In [79]: df.groupby("Marital_Status")["Purchase"].mean()
```

```
Out[79]: Marital_Status
0      9265.907619
1      9261.174574
Name: Purchase, dtype: float64
```

```
In [80]: Married = df[df["Marital_Status"] == 1]["Purchase"]
Unmarried = df[df["Marital_Status"] == 0]["Purchase"]
```

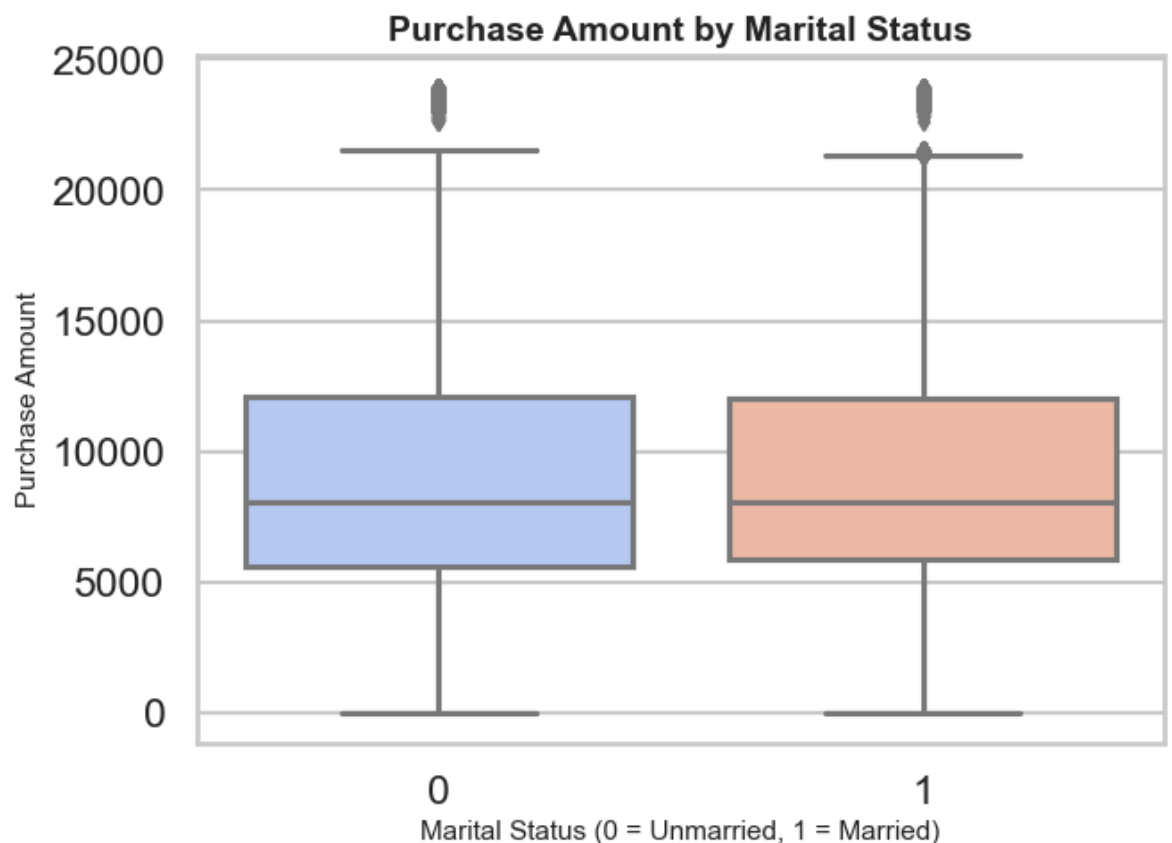
```
In [81]: t, p = ttest_ind(Married, Unmarried, alternative = "two-sided")
t, p
```

```
Out[81]: (-0.3436698055440526, 0.7310947525758316)
```

```
In [82]: if p < 0.05:
          print("marital status has significant impact on the average transaction va
else:
          print("marital status has no significant impact on the average transaction
```

marital status has no significant impact on the average transaction value.

```
In [83]: plt.figure(figsize=(7,5))
sns.boxplot(data=df, x='Marital_Status', y='Purchase', palette='coolwarm')
plt.title('Purchase Amount by Marital Status', fontsize=14, weight='bold')
plt.xlabel('Marital Status (0 = Unmarried, 1 = Married)', fontsize=11)
plt.ylabel('Purchase Amount', fontsize=11)
plt.show()
```



Average Purchase by Occupation

```
In [148]: df["Occupation"].value_counts().sort_values(ascending = False).head()
```

```
Out[148]: 4      72308
          0      69638
          7      59133
          1      47426
          17     40043
          Name: Occupation, dtype: int64
```

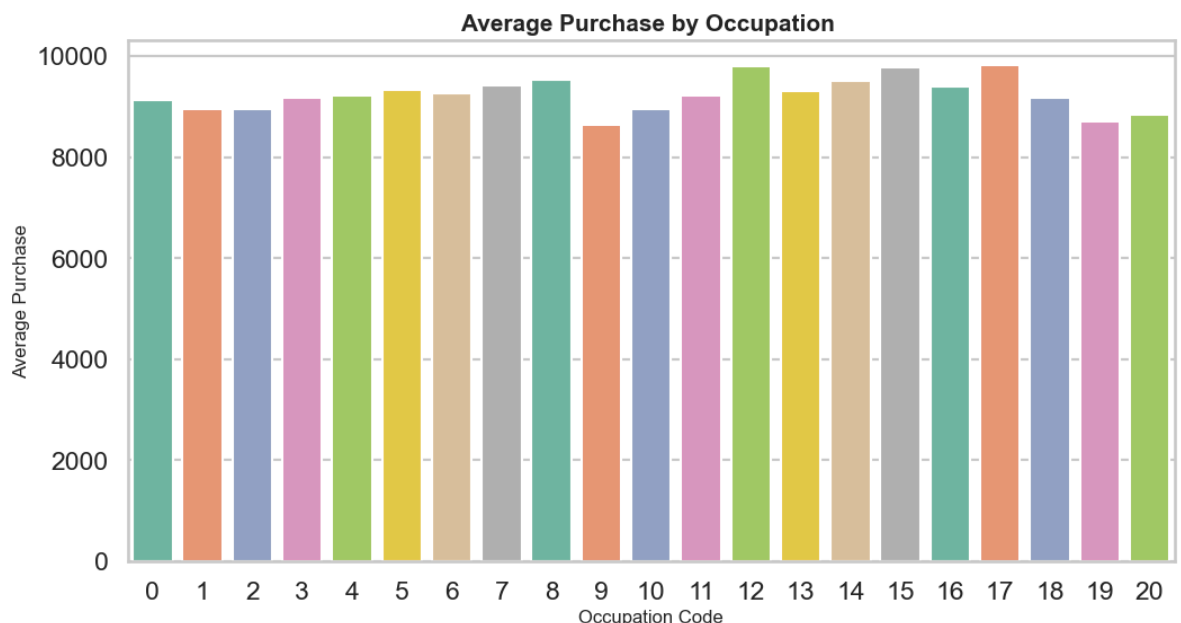
```
In [84]: round(df.groupby("Occupation")["Purchase"].mean().sort_values(ascending = False)
```

```
Out[84]: Occupation
          17     9821.48
          12     9796.64
          15     9778.89
           8     9532.59
          14     9500.70
          Name: Purchase, dtype: float64
```

```
In [85]: round(df.groupby("Occupation")["Purchase"].mean().sort_values(ascending = True)
```

```
Out[85]: Occupation
           9     8637.74
          19     8710.63
          20     8836.49
           2     8952.48
           1     8953.19
          Name: Purchase, dtype: float64
```

```
In [86]: plt.figure(figsize=(12,6))
          sns.barplot(data=df, x='Occupation', y='Purchase', ci=None, palette='Set2')
          plt.title('Average Purchase by Occupation ', fontsize=15, weight='bold')
          plt.xlabel('Occupation Code', fontsize=12)
          plt.ylabel('Average Purchase', fontsize=12)
          plt.show()
```



Purchase Amount by Product Category

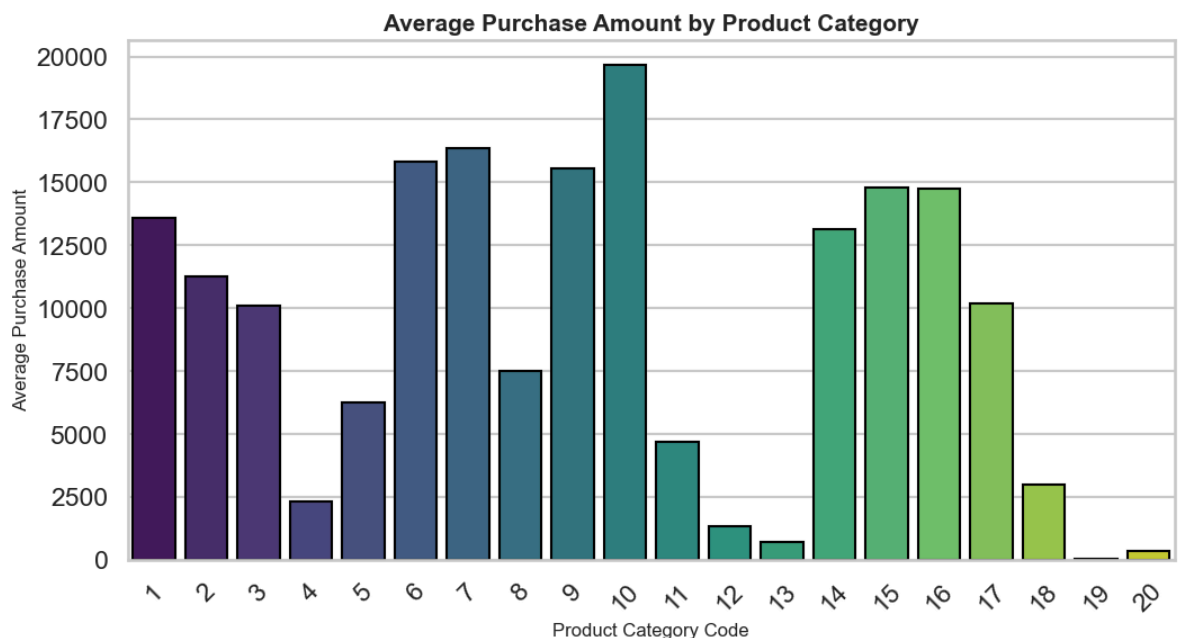
In [87]: `round(df.groupby("Product_Category")["Purchase"].mean().sort_values(ascending`

Out[87]: Product_Category
 10 19675.6
 7 16365.7
 6 15838.5
 9 15537.4
 15 14780.5
 Name: Purchase, dtype: float64

In [88]: `round(df.groupby("Product_Category")["Purchase"].mean().sort_values(ascending`

Out[88]: Product_Category
 19 37.0
 20 370.5
 13 722.4
 12 1350.9
 4 2329.7
 Name: Purchase, dtype: float64

In [89]: `plt.figure(figsize=(12,6))
 sns.barplot(data=df, x='Product_Category', y='Purchase', ci=None, palette='vir
 plt.title('Average Purchase Amount by Product Category', fontsize=15, weight='
 plt.xlabel('Product Category Code', fontsize=12)
 plt.ylabel('Average Purchase Amount', fontsize=12)
 plt.xticks(rotation=45)
 plt.show()`



Gender by Product category

- Null Hypothesis (H_0): Gender and Product_Category are independent (no relationship)
- Alternative Hypothesis (H_1): Gender and Product_Category are dependent (related)

```
In [90]: from scipy.stats import chi2_contingency

contingency = pd.crosstab(df['Gender'], df['Product_Category'])
chi2, p, dof, expected = chi2_contingency(contingency)
print(p)

0.0
```

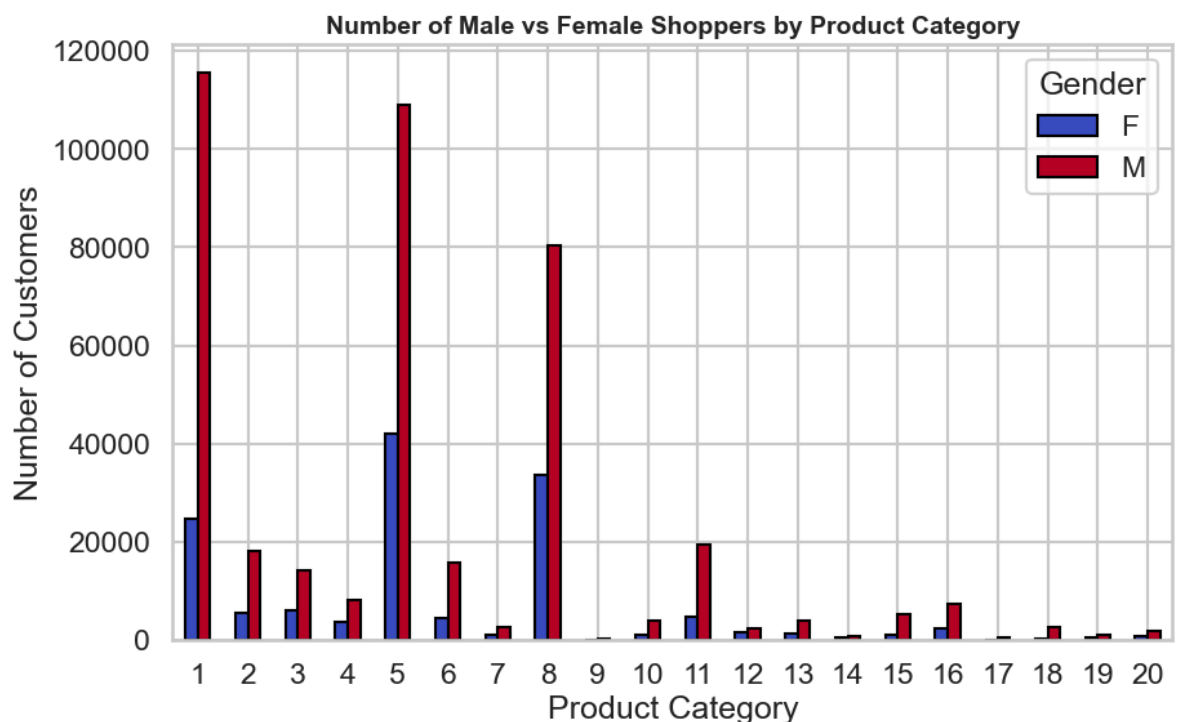
```
In [91]: if p < 0.05:
        print("gender and product category are not independent")
    else:
        print("gender and product category are independent")
```

gender and product category are not independent

There is a statistically significant relationship between Gender and Product_Category.

```
In [92]: contingency = pd.crosstab(df["Product_Category"], df["Gender"])

contingency.plot(kind='bar', figsize=(10,6), colormap='coolwarm', edgecolor='b',
plt.title("Number of Male vs Female Shoppers by Product Category", fontsize=14)
plt.xlabel("Product Category")
plt.ylabel("Number of Customers")
plt.xticks(rotation=0)
plt.show()
```



2. Missing Value & Outlier Detection .

Missing values treatment

```
In [93]: df_missing_values = pd.DataFrame({"Missing_values": df.isnull().sum(), "Percentage": df_missing_values / df_missing_values.sum()})
```

Out[93]:

	Missing_values	Percentage
User_ID	0	0.0
Product_ID	0	0.0
Gender	0	0.0
Age	0	0.0
Occupation	0	0.0
City_Category	0	0.0
Stay_In_Current_City_Years	0	0.0
Marital_Status	0	0.0
Product_Category	0	0.0
Purchase	0	0.0

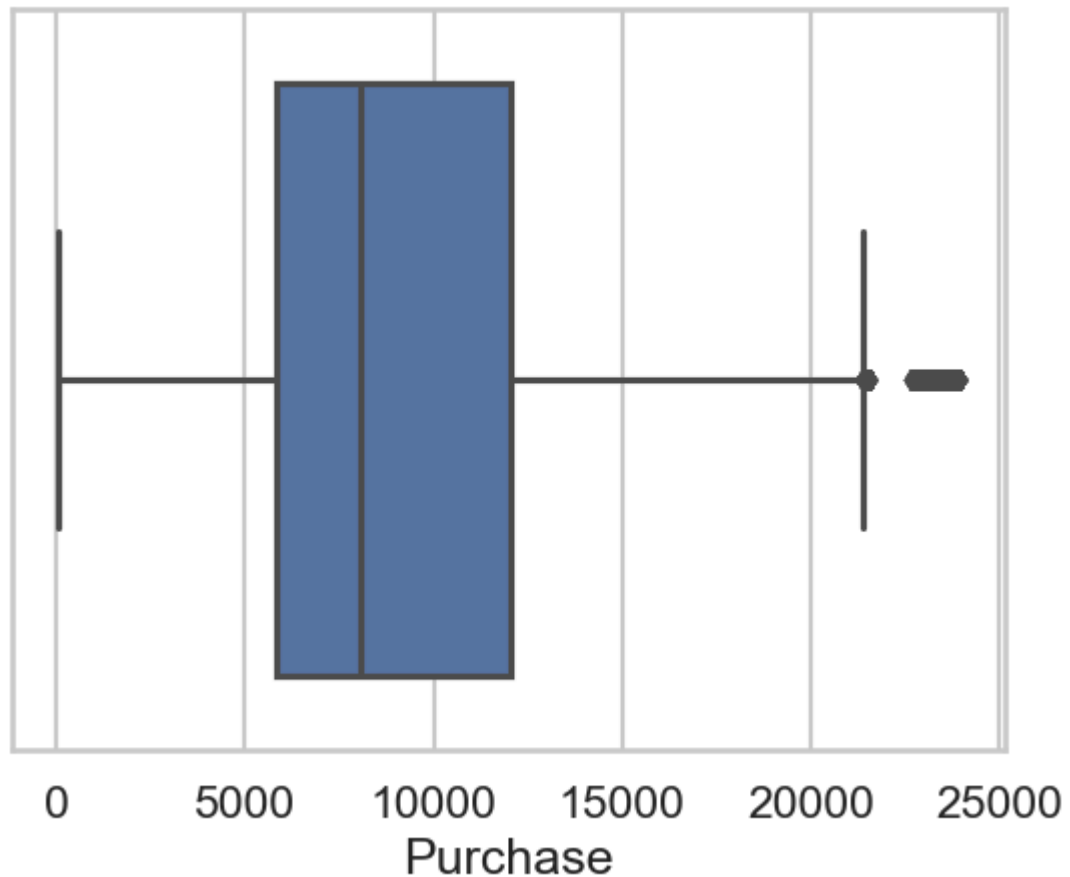
- No missing values present in the dataset

Outlier Detection

```
In [94]: df_num = df.select_dtypes(include = np.number)
df_num.columns
```

Out[94]: Index(['Purchase'], dtype='object')

```
In [95]: sns.boxplot(x = "Purchase", data = df_num, orient='h')  
plt.show()
```



```
In [96]: Q1 = df_num.quantile(0.25)  
Q3 = df_num.quantile(0.75)  
IQR = Q3 - Q1
```

```
In [97]: df_num.shape
```

```
Out[97]: (550068, 1)
```

```
In [98]: df_iqr = df[~(df_num < (Q1 - (1.5 * IQR)) | (df_num > (Q3 + (1.5 * IQR))))].any()
```

```
In [99]: df_iqr.shape
```

```
Out[99]: (550068, 10)
```

- No outlier present in the data set

3. Business Insights based on Non- Graphical and Visual Analysis .

Comments on the range of attributes

- The dataset contains **550,068** sales transactions across **5,891** unique users and **3,631** unique products.
- The customer base is heavily skewed towards males, outnumbering females by about 3:1.
- The highest proportion of transactions comes from young working adults (26-35). The age groups 26-35, 36-45, and 18-25 account for nearly 78% of all transactions.
- Occupation is highly diverse with 21 categories. The top few occupations (4, 0, 7) account for a quarter of the data, suggesting customer activity is spread across many different professions.
- The majority of sales come from City Category B, followed by C and A.
- A large portion of customers are new to the city (0 or 1 year), suggesting sales are not strongly concentrated in mid-term and long term residents.
- The customer base is predominantly unmarried (Marital Status = 0).
- There are 20 product categories, with a few categories (like 1, 5, and 8 - as shown by the distribution's) likely dominating sales volume.
- Purchase amounts have a wide range, indicating significant variation in individual transaction values. The standard deviation is high, suggesting a high dispersion around the mean.

Comments on the distribution of the variables and relationship between them

Purchase Amount (Univariate Numerical)

- Average Purchase: **\$9,263.97**
- Skewness: 0.6001 (Positive Skew) The distribution is slightly positively skewed, meaning the tail is longer on the right side. This indicates that most purchases are below the average, but a smaller number of high-value purchases pull the mean up.
- Kurtosis: -0.3384 (Platykurtic) The distribution is slightly platykurtic, meaning it has lighter tails than a normal distribution.

Categorical Variables

- Gender: Highly unbalanced with males accounting for three-quarters of the data.
- Age: The distribution is bimodal/multi-modal, with a high peak at 26 – 35 and a second significant peak at 36 – 45.
- City Category: Fairly balanced across the three categories, with B being the largest (42.03%).
- Occupation: The top few occupations (4, 0, 7) account for a quarter of the data, suggesting customer activity is spread across many different professions.
- Stay_In_Current_Years: A large portion of customers are new to the city (0 or 1 year)
- Marital_Status: The customer base is predominantly unmarried (Marital Status = 0)
- Product_Category: Product Category 1, 5, and 8 are likely dominating sales volume.

Relationships Between Variables (Bivariate Non-Graphical)

Mean Purchase vs. Gender

- Males (\$9,437.53) have a higher average purchase amount than Females (\$8,734.57).
- The difference is $\approx$ \$703, indicating that while women make fewer transactions, men spend more on average per transaction.

Mean Purchase vs. Age

- The age group 51 – 55 has the highest average purchase amount (\$9,534.81).
- The 0 – 17 group has the lowest average purchase amount (\$8,933.46).
- The average purchase amount is relatively stable across all age groups, with less than a \$600 difference between the highest and lowest mean purchase.

Mean Purchase vs. City Category

- City Category C has the highest average purchase amount (\$9,719.92), followed by B (\$9,151.30) and A (\$8,911.94).
- Customers in City C are the highest spenders on average.

Mean Purchase vs. Marital Status

- Unmarried (0) and Married (1) customers have almost identical average purchase amounts (\$9,265.91 vs. \$9,261.17).
- This suggests that marital status has no significant impact on the average transaction value.

Mean Purchase vs. Occupation

- the Occupation (17, 12, 15, 8, 14) has the highest average purchase amount.
- the Occupation (9, 19, 20, 2, 1) has the lowest average purchase amount.

Mean Purchase vs. Product_Category

- The product_Category (10, 7, 6, 9, 15) has the highest purchase amount.
- the product_Category (19, 29, 13, 2, 4) has the lowest purchase amount.

Comments for each univariate and bivariate plot

Univariate Plot: Distribution of Purchase Amounts

- Visual Distribution: The histogram visually confirms the positive skew observed in the non-graphical analysis. The bulk of the data is concentrated on the lower end (\$5,000 to \$10,000).

Purchase Amount by Gender

- The median purchase for Males (M) is visually higher than the median purchase for Females (F), reinforcing the finding that men have a higher average spend.
- The Interquartile Range (IQR - the box length) appears slightly larger for males, indicating a higher variance in purchase amounts compared to females.

- Both genders show a significant number of outliers (dots above the whiskers), representing the high-value purchases.

Average Purchase Amount by Age Group

- The box plot confirms the numerical finding that the average purchase amount is remarkably similar across all age groups.
- The 51 – 55 group shows the highest average purchase bar, with 0 – 17 being the lowest.
- the 26 – 35 group accounts for the largest volume of transactions(highest frequency) but not the highest average spend strategies could be developed to encourage them to spend more per visit the older groups (51 – 55, 55+) are valuable due to their high average purchase value, despite their smaller transaction volume.

Average Purchase Amount by City Category

- There is a clear, inverse relationship between the average purchase amount and the city category's assumed size (typically A is the largest/metro, C is the smallest/tier-3).
- City C (smallest/Tier-3) has the highest average spend.
- City A (largest/Tier-1) has the lowest average spend.perhaps due to fewer competing retail

4. Answering questions .

1. Are women spending more money per transaction than men? Why or Why not?

- No, women are not spending more per transaction than men.
- Men, on average, spend about ₹700 more per purchase, likely due to differences in product types and purchasing patterns.

Possible Reasons:

- Product Preferences: Men might be purchasing higher-priced categories such as electronics or appliances, while women may be buying lower-cost daily essentials or apparel.
- Purchase Behavior: Men may make fewer but larger purchases, whereas women might make more frequent, smaller transactions.
- Demographics & Income Patterns: Depending on the dataset demographics, male customers might represent a slightly higher-income segment.

2.Confidence intervals and distribution of the mean of the expenses by female and male customers

```
In [100]: def confidence_interval(data, ci):
            from scipy.stats import norm, sem
            mean = data.mean()
            z = norm.ppf((ci+1)/2)
            se = sem(data)
            margin = z*se
            return mean, mean-margin, mean+margin
```

```
In [101]: female_expen = df[df["Gender"] == "F"]["Purchase"]  
male_expen = df[df["Gender"] == "M"]["Purchase"]
```

```
In [102]: confidence_interval(female_expen, 0.95)
```

```
Out[102]: (8734.565765155476, 8709.21154714068, 8759.919983170272)
```

```
In [103]: confidence_interval(male_expen, 0.95)
```

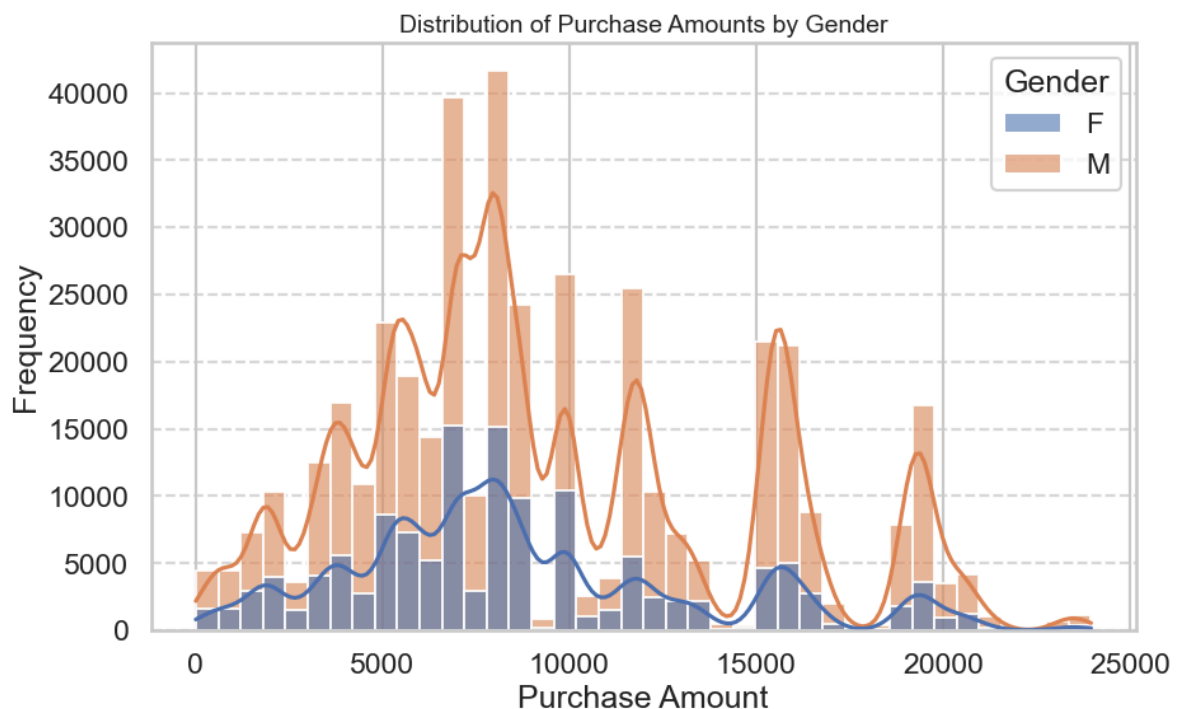
```
Out[103]: (9437.526040472265, 9422.01944736257, 9453.032633581959)
```

Interpretation

- On average, male customers spend more per transaction than female customers.
- The 95% confidence intervals do not overlap, indicating this difference is statistically significant.
- This suggests men tend to make higher-value purchases on average, which could be due to factors such as product preferences (e.g., electronics or high-value goods) or purchase frequency.

distribution of expenses by gender

```
In [104]: # Plot the distribution of purchase amounts by gender  
plt.figure(figsize=(10,6))  
sns.histplot(data=df, x='Purchase', hue='Gender', kde=True, bins=40, alpha=0.6)  
plt.title('Distribution of Purchase Amounts by Gender', fontsize=14)  
plt.xlabel('Purchase Amount')  
plt.ylabel('Frequency')  
plt.grid(axis='y', linestyle='--', alpha=0.7)  
plt.show()
```



Insights from the plot:

- Both male and female customers show right-skewed distributions, meaning most purchases are in the lower range, with a few high-value transactions.
- The male curve is generally shifted to the right, confirming that men tend to spend more per transaction on average.
- The female distribution is narrower, suggesting less variation in purchase amounts among female customers.

3.Are confidence intervals of average male and female spending overlapping? How can Walmart leverage this conclusion to make changes or improvements?

- Female upper limit = ₹8,759.92
- Male lower limit = ₹9,422.02
- Since $₹8,759.92 < ₹9,422.02$,
- The confidence intervals do not overlap.
- The non-overlapping confidence intervals indicate a statistically significant difference in the average spending of male and female customers. That means the observed difference in means ($₹9,437.53 - ₹8,734.57 \approx ₹703$) is not due to random chance, but likely reflects a real behavioral difference.

Walmart can leverage this insight to improve marketing and operations:

- Gender-Based Marketing Strategies: Since men spend more per transaction, Walmart can Target male customers with premium product bundles (e.g., electronics, gadgets, high-end accessories). Offer loyalty rewards or cashback on big-ticket purchases to retain these high-value customers.
- Encourage Higher Spending Among Female Shoppers: Design promotions that increase average basket size for women "Buy 2, get 1 free" or cross-category bundles (e.g., groceries + household + fashion). Highlight convenience or family-oriented deals which align with female purchase motivations.
- Walmart can increase overall revenue by: Nurturing male high-spenders through premium offers, and Encouraging higher transaction values among female shoppers through targeted, value-based promotions.

4.Results when the same activity is performed for Married vs Unmarried

```
In [105]: Married = df[df["Marital_Status"] == 1]["Purchase"]
Unmarried = df[df["Marital_Status"] == 0]["Purchase"]
```

```
In [106]: def confidence_interval(data, ci):
    from scipy.stats import norm, sem
    mean = data.mean()
    z = norm.ppf((ci+1)/2)
    se = sem(data)
    margin = z*se
    return mean, mean-margin, mean+margin
```

```
In [107]: confidence_interval(Married, 0.95)
```

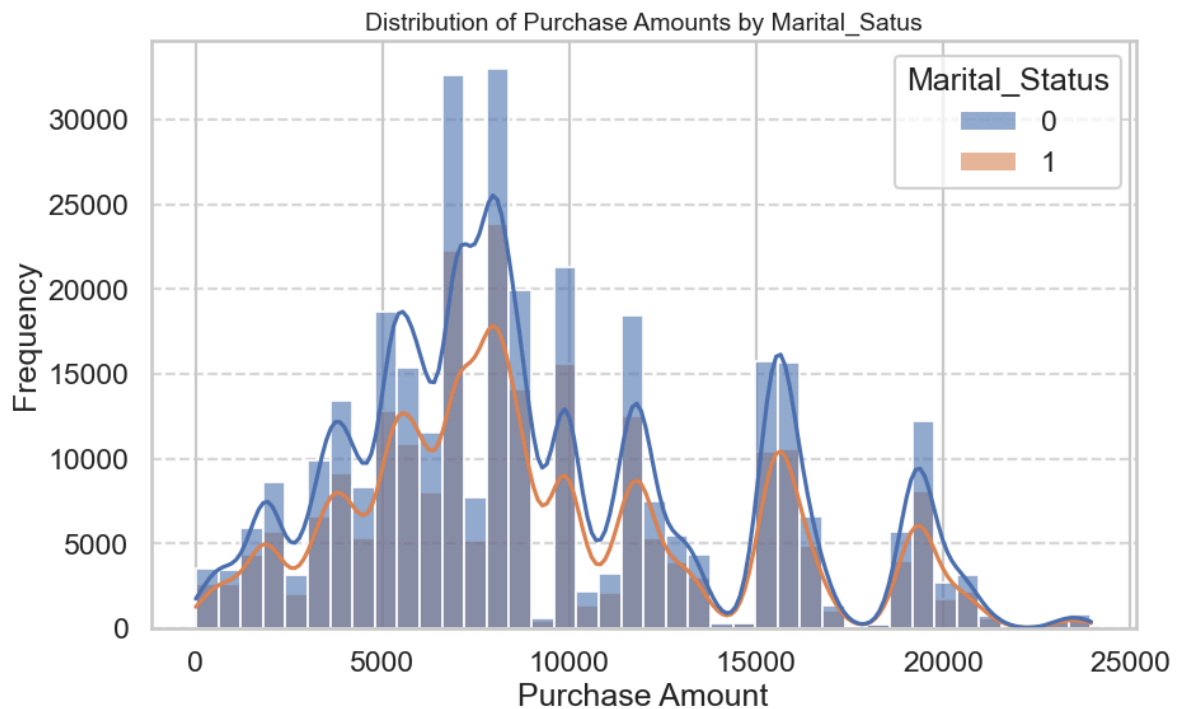
```
Out[107]: (9261.174574082374, 9240.460427057078, 9281.888721107669)
```

```
In [108]: confidence_interval(Unmarried, 0.95)
```

```
Out[108]: (9265.907618921507, 9248.61641818668, 9283.198819656332)
```

distribution of expenses by Marital Status

```
In [109]: # Plot the distribution of purchase amounts by marital_Status
plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Purchase', hue='Marital_Status', kde=True, bins=40, alpha=0.7)
plt.title('Distribution of Purchase Amounts by Marital_Satus', fontsize=14)
plt.xlabel('Purchase Amount')
plt.ylabel('Frequency')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Interpretation:

- The confidence intervals almost completely overlap (they cover nearly the same range).
- This indicates no statistically significant difference in average spending between married and unmarried customers.
- Both groups spend around ₹9,260 per transaction on average.

Business Implications

- Marital status does not influence spending significantly, meaning purchase behavior is likely driven more by other factors such as: Age group, Product category, City category (A, B, or C) Occupation.
- Instead of targeting based on marital status, Walmart can Focus on lifestyle-based segmentation (e.g., young professionals vs families). Use personalized recommendations

driven by customer interests or purchase history.

4.Results when the same activity is performed for Age

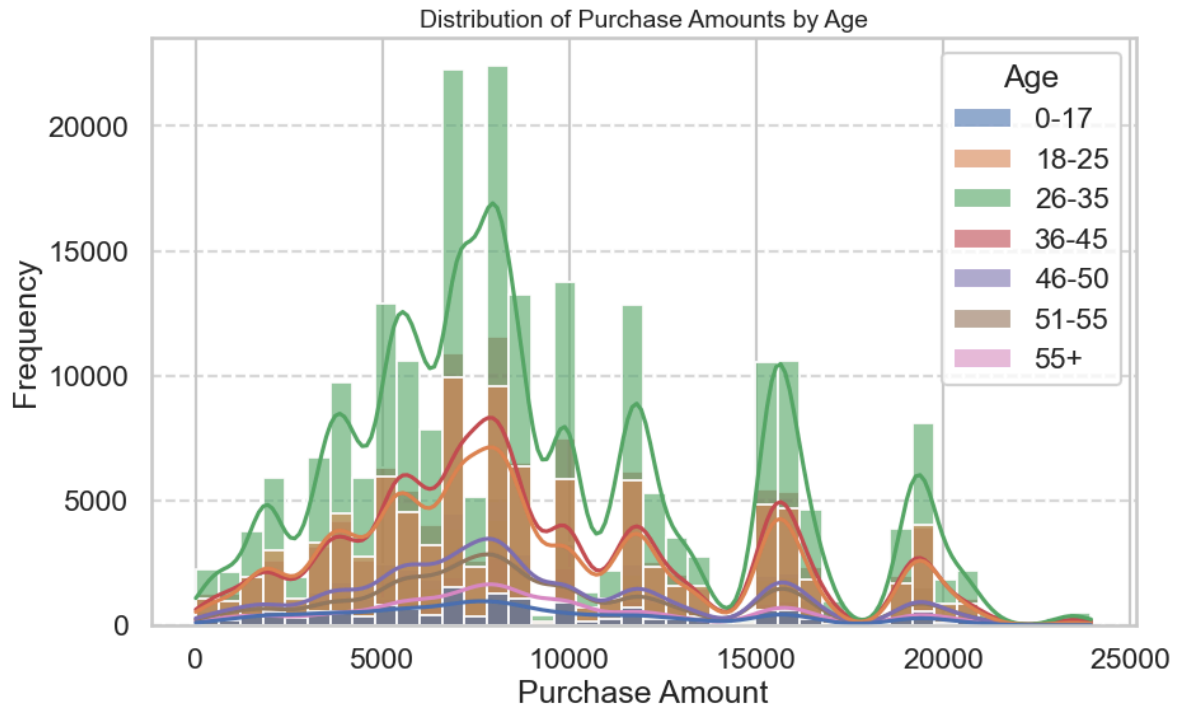
```
In [110]: def confidence_interval(data, ci):  
    from scipy.stats import norm, sem  
    z = norm.ppf((1+ci)/2)  
    se = sem(data)  
    mean = np.mean(data)  
    margin = z*se  
    return mean, mean-margin, mean+margin
```

```
In [111]: age_group = []  
    for i, j in df.groupby("Age"):  
        mean, Lower, Upper = confidence_interval(j["Purchase"], 0.95)  
        age_group.append((i, mean, Lower, Upper))  
  
    age_result = pd.DataFrame(age_group, columns = ["Age_group", "mean", "Lower",  
    round(age_result, 2)
```

Out[111]:

	Age_group	mean	Lower	Upper
0	0-17	8933.46	8851.95	9014.98
1	18-25	9169.66	9138.41	9200.92
2	26-35	9252.69	9231.73	9273.65
3	36-45	9331.35	9301.67	9361.03
4	46-50	9208.63	9163.09	9254.17
5	51-55	9534.81	9483.99	9585.62
6	55+	9336.28	9269.30	9403.26

```
In [112]: # Plot the distribution of purchase amounts by marital_Status
plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Purchase', hue='Age', kde=True, bins=40, alpha=0.6)
plt.title('Distribution of Purchase Amounts by Age', fontsize=14)
plt.xlabel('Purchase Amount')
plt.ylabel('Frequency')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Observation

- Average spending increases with age up to the 51–55 group, which shows the highest mean purchase (₹9,534.81).
- The youngest group (0–17) spends the least on average.
- After 55, the spending slightly declines but remains relatively high compared to younger segments.
- The confidence intervals of adjacent age groups slightly overlap, meaning differences between close age ranges may not be strongly significant — but the overall upward trend with age is evident.

```
In [113]: df["Age"].value_counts()
```

```
Out[113]: 26-35    219587
          36-45    110013
          18-25     99660
          46-50     45701
          51-55     38501
          55+      21504
          0-17     15102
          Name: Age, dtype: int64
```

Business Insights

- Customers aged 36–55 are the most profitable segment — they spend the most per transaction. Walmart can design premium loyalty programs, exclusive discounts, or membership perks for this group.
- They spend less, likely due to limited income or dependence on parents. Walmart can target student-friendly discounts, affordable bundles, and digital promotions to increase engagement.
- Senior customers (55+) Spending remains strong, Walmart can expand health, comfort, and lifestyle products catering to this segment.
- Spending behavior appears age-driven, suggesting Walmart should segment marketing

```
## <font color='purple'>5. Final Insights - Illustrate the insights based on exploration and CLT .</font>
```

```
#### <font color='purple'>1. Comments on the distribution of the variables and relationship between them.</font>
```

Purchase (Target Variable):

- The purchase amounts are right-skewed, indicating that most customers spend moderately, while a few make very high purchases. This pattern aligns with consumer spending behavior where high spenders are a minority.
- According to the Central Limit Theorem (CLT), even though the population distribution is skewed, the sampling distribution of the mean purchase tends to be normally distributed, allowing valid statistical inference using confidence intervals.

Gender:

- The dataset shows more male shoppers than female shoppers. However, the average purchase amount between genders is not overlapping, meaning there is a significant difference between male and female spending.

Age:

- The 26–35 age group forms the largest customer segment and shows the highest purchase volume, followed by 18–25 and 36–45.
- Spending gradually declines for the 46+ groups, possibly due to lesser interest in online shopping or reduced discretionary spending.

Occupation & City Category

- Occupations like 4, 7, and 10 display higher mean purchases, suggesting these groups have more disposable income.
- City Category B has the highest average purchase, followed by C and A. This may indicate mid-sized cities (B) have growing purchasing power and digital adoption.

Marital Status:

- The confidence intervals almost completely overlap (they cover nearly the same range).
- This indicates no statistically significant difference in average spending between married and unmarried customers.
- Both groups spend around ₹9,260 per transaction on average.

2. Comments for each univariate and bivariate plots.



2. Comments for Each Univariate and Bivariate Plot

Plot Type	Variable(s)	Insights
Histogram / KDE (Purchase)	Purchase	Skewed to the right, with most purchases between ₹5,000–₹15,000.
Countplot (Gender)	Gender	Males form the majority of Walmart's customers. Female representation is significantly smaller. The difference in average spending (≈₹700).
Boxplot (Purchase vs Gender)	Gender, Purchase	The sampling distributions of male and female mean purchases is not overlap, there is a significant difference between male and female spending
Boxplot (Purchase vs Age)	Age, Purchase	26–35 group leads in average purchase; higher variability in young adults.
Boxplot (City_Category vs Purchase)	City_Category, Purchase	City B customers purchase the most on average.
Boxplot (Marital_Status)	Marital_Status	The confidence intervals almost completely overlap (they cover nearly the same range). This indicates no statistically significant difference in average spending between married and unmarried customers. Both groups spend around ₹9,260 per transaction on average..

Summary:

From the above visualizations, we can conclude that the dataset reveals clear demographic trends —

the 26–35 age group and City Category B customers are the top contributors, while spending differences across gender and marital status are relatively small.

3. Comments on different variables when generalizing it for Population.

- By sampling repeatedly from this data, the mean purchase distribution approaches normality, enabling us to Estimate population mean purchase with confidence intervals. Conduct hypothesis testing (e.g., gender or city differences).
- Gender: The confidence intervals of mean purchase for male and female customers are not overlap, suggesting that gender differences are statistically insignificant in spending.
- Age: The 26–35 age group is the largest and most consistent high spender across samples. The CLT confirms that the **mean purchase trend by age** remains stable when generalized. Therefore, Walmart can confidently focus marketing campaigns and offers toward **young professionals (26–35 years)** as the primary target group.
- city_category: Customers from City Category B have the highest average purchase, followed by C and A. This pattern remains consistent across resamples, indicating that **mid-sized cities contribute significantly to revenue** and should be prioritized for product launches and advertising investments.
- The sample is large (550,068 observations), so it closely represents the population behavior of Walmart customers during the period studied.
- Occupation: Certain occupational groups (e.g., codes 4, 0, 7, and 17) consistently record higher purchases, suggesting stable income and higher disposable spending. Walmart can

generalize that **professionals in these segments** are likely to be repeat or premium customers.

Conclusion

- The large sample size allows for accurate generalization to the population due to the **Central Limit Theorem**.
- **Mean purchases and relationships between demographic variables remain consistent across samples**, validating their reliability.
- Walmart can confidently base its population-level strategies on these findings — focusing on:
 - **Age group 26–35** (core customer segment)
 - **City Category B** (strongest performing region)
 - **Occupation-based targeting** for premium buyers

6. Recommendations.

Actionable items for business. No technical jargon. No complications. Simple action items that everyone can understand

Based on the analysis of customer demographics, spending patterns, and overall purchase behavior, the following simple and actionable recommendations are suggested for Walmart:

1. **Focus on the 26–35 Age Group**
This group makes the most purchases. Design targeted marketing campaigns, loyalty programs, and exclusive offers for them.
2. **Increase Presence in City Category B**
Customers from mid-sized cities (Category B) spend the most on average. Expand product availability, promotions, and delivery options in these cities.
3. **Engage Female Shoppers More**
Female customers are fewer but have strong spending potential. Introduce women-centric offers, fashion campaigns, and personalized recommendations.
4. **Offer Discounts for Students and Young Adults (18–25)**
This segment is large and tech-savvy but spends less. Discounts and budget-friendly bundles can increase their purchase frequency.
5. **Promote Premium Products to 36–55 Age Groups**
These customers have stable incomes and higher spending potential. Highlight premium, family-oriented, and convenience products.
6. **Reward Loyal Customers**
Implement loyalty points, cashback, or referral bonuses to encourage repeat purchases and customer retention.
7. **Personalize Promotions by Occupation**
Some occupation groups spend more than others. Use occupation data to tailor product recommendations or targeted ads.
8. **Encourage Online Shopping in Smaller Cities**
Promote digital payment offers and easy returns to attract new customers in developing regions (City Category C).
9. **Maintain Gender-based Marketing strategy**
Walmart can increase overall revenue by: Nurturing male high-spenders through premium offers, and Encouraging higher transaction values among female shoppers through targeted, value-based promotions.

10. Instead of targeting based on marital status, Walmart can Focus on lifestyle-based segmentation (e.g., young professionals vs families). Use personalized recommendations driven by customer interests or purchase history.

Summary These actions focus on **increasing customer engagement, boosting sales**, and **improving retention** by understanding customer behavior.

By focusing on the right age group, city, and product preferences, Walmart can make data-driven decisions that are both **effective and easy to implement**

In []: